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TITLE PAGE

- **Problem Statement ID –** Open Innovation
- **Problem Statement Title-**

ITC RADAR: AI-Powered Input Tax Credit Risk Detection

- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)**

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ITC RADAR: AI-Powered Input Tax Credit Risk Detection

From Declarations to Ground Reality, From Suspicion to Stronger Revenue

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> PROPOSED SOLUTION: ITC RADAR

An AI-assisted risk detection platform that combines digital GST data with independent geospatial, market and network evidence to identify suspicious ITC refund patterns and support human investigation.

- **Our Vision:** A transparent, data-driven system that protects government revenue by detecting fraudulent ITC claims using **physical, economic and network evidence**.
- **What it does:** Integrates GST/business data, satellite & geospatial data, trade benchmarks and invoice networks, and analyses them using AI/ML to generate risk scores, alerts and explainable investigation dossiers.
- **Why it is different:** Goes beyond document-based checks by validating whether the declared business activity is **physically plausible** and **economically reasonable** using independent third-party data sources.

GST / Business Data
Invoices | E-way Bills | HSN
ITC | Refund Claims

Geospatial & Satellite Data
Overture Buildings | ESA WorldCover
OpenStreetMap | Sentinel-1/2

Trade & Market Data
UN Comtrade | Govt. Trade Statistics
HSN Price Benchmarks

Network / Graph Data
Supplier-Recipient Links
Shared Identifiers | Circular Trades

ITC RADAR
AI DETECTION
RISK SCORING
EVIDENCE FUSION

Risk Alerts
High / Medium / Low Risk

Investigation Dossier
Maps | Calculations
Network Graphs | Evidence

Human Verification
Supports Officers
Does not auto-declare fraud

Physical + Market + Network Evidence → Explainable Insights → Stronger Revenue

> KEY FEATURES & UNIQUE SELLING PROPOSITION (USP):

1. Real-World Verification (Gerspatial Intelligence)
- Uses satellite imagery and building footprints to verify if the registered premises can support the declared business activity.
- Detects dummy or non-operational entities.

2. Economic Plausibility (Price Closure)
- Compares declared unit prices with HSN-level market benchmarks (UN Comtrade, govt. trade data).
- Flags over-invoicing and abnormal pricing patterns.

3. Premises Aggregation (Capacity Closure)
- Identifies multiple GSTINs sharing the same or nearby premises.
- Estimates physical handling capacity and tests declared throughput.

4. Network Topology Analysis
- Detects suspicious ITC chains, circular trading and weak tax origin.
- Uses graph analytics to identify connected fraud networks.

5. Explainable & Transparent
- Every risk flag is traceable to source data, calculation and named detector.
- Generates an evidence dossier with maps, charts and summaries.

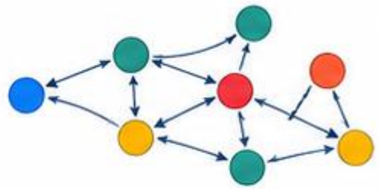
6. Human-in-the-Loop
- Provides risk scores and evidence to tax officers for further investigation.
- Does not automatically declare fraud or trigger financial action.

Satellite View (Overture + WorldCover)

Premises Analysis

Registered Premises	1,280 m²
Land Cover	Industrial
Road Access	Yes
Nearby GSTINs	4

Invoice Network (Sample)



Supplier
Recipient
High Risk
Shared Premises

ITC RADAR

Dashboard
Risk Alerts 12
Entity Search
Network Graph
Map View
Reports

Total Entities Analysed	High Risk	Medium Risk	Low Risk
12,450	318	1,246	10,886

Risk Distribution

High 2.6%
Medium 10.0%
Low 87.4%

Risk Map (India)

High
Medium
Low

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TECHNICAL APPROACH

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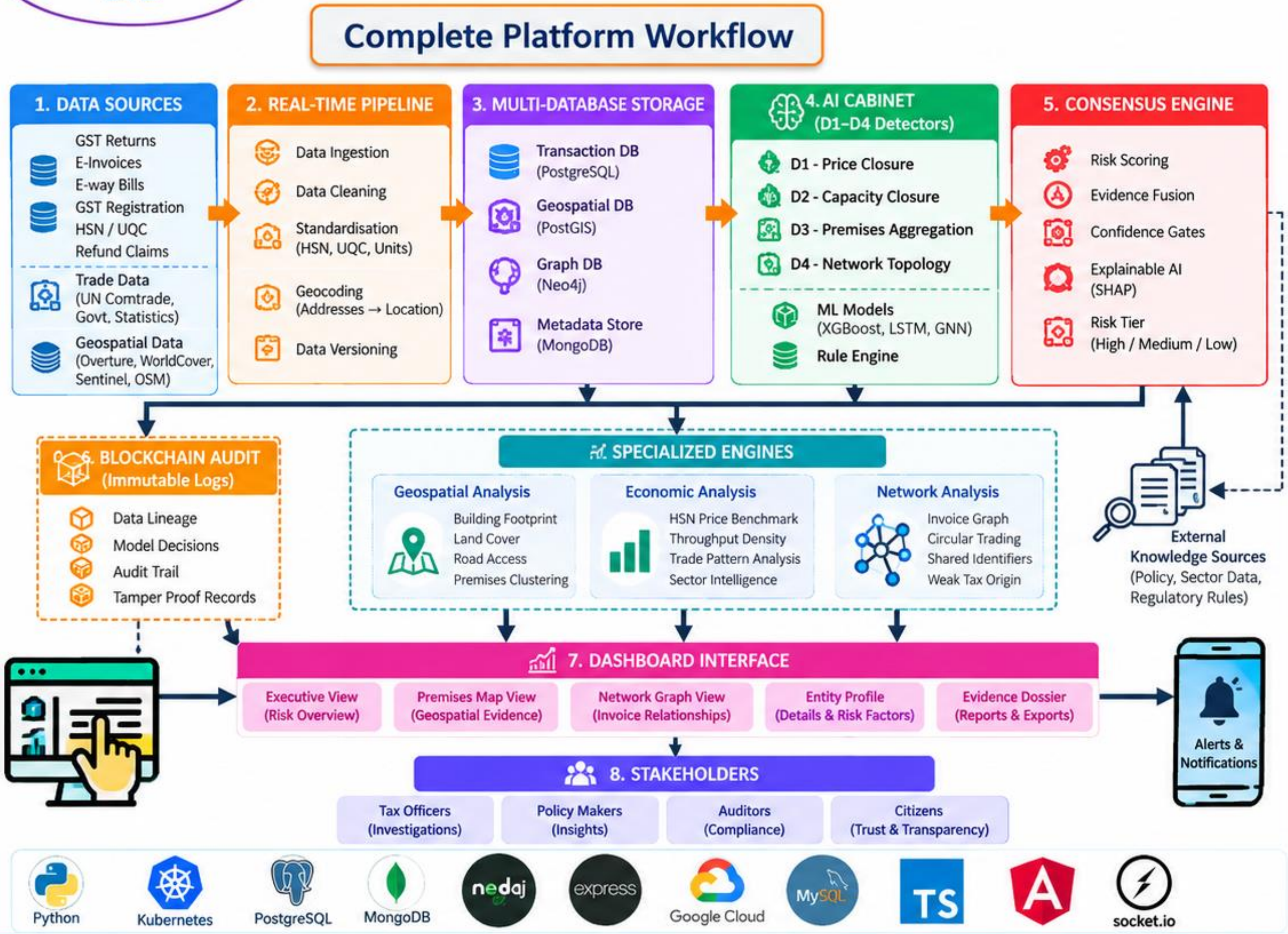
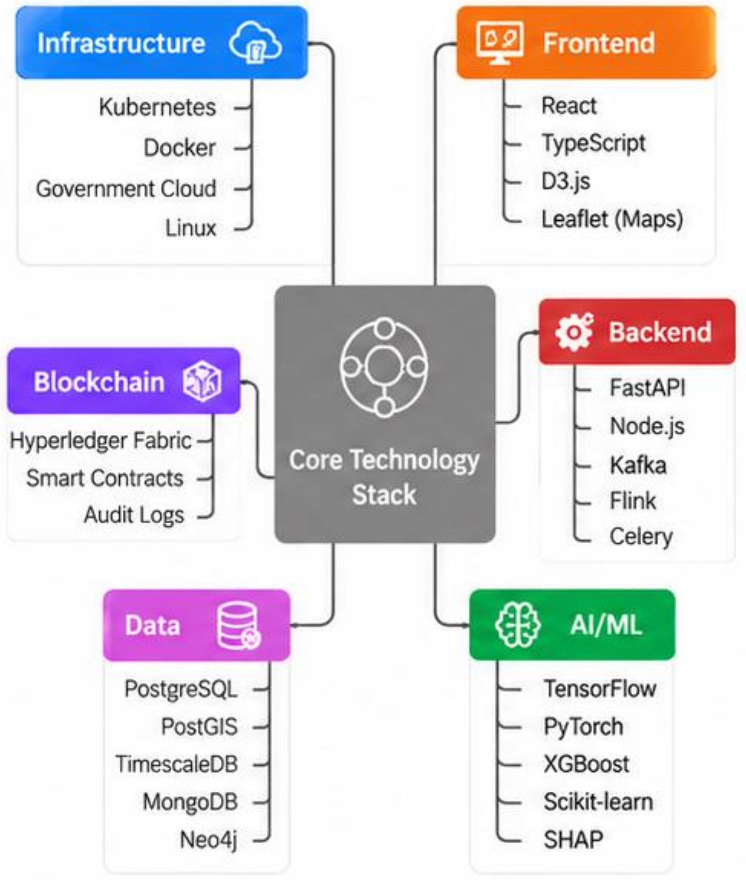


Core Technology Stack Overview



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FEASIBILITY AND VIABILITY

Practical • Scalable • Secure • High-Impact for Tax Administration

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FEASIBILITY OF ITC RADAR

MVP-Ready
- Built using open-source tools and dataset (Overture Buildings, ESA WorldCover, OSM).
- Uses synthetic GST data for prototype.
- Modular design with D1–D4 detectors.

Explainable
- Each risk flag is traceable to data source, calculation and named detector.
- Maps, charts and network graphs for easy understanding.
- Transparent reasoning using SHAP-style insights.

Privacy-Safe
- No sensitive taxpayer data in prototype (synthetic data only).
- Role-based access and audit logs.
- Designed with data minimisation principles.

Extensible
- New HSN sectors, detectors and data sources can be added easily.
- Supports regional languages and wider geospatial coverage.
- Can integrate with government systems later.

Scalable & Efficient
- Cloud-native and containerized deployment.
- Region-wise / state-wise processing.
- Designed to handle large volumes of GSTINs and transactions.

POTENTIAL CHALLENGES AND STRATEGIES FOR OVERCOMING

Key Challenges
- Inaccurate or incomplete addresses.
- Missing or low-quality building footprints.
- Sector-wise variation in throughput and unit prices.
- Risk of false positives.
- Limited availability of real transaction data.

Mitigation Strategy
- Use multiple geocoding sources and confidence scores.
- Combine Overture, ESA WorldCover, Sentinel and OSM.
- Use sector-aware benchmarks and robust thresholds.
- Apply role and confidence gates with corroboration across D1–D4.
- Develop and test with synthetic data first; integrate real data in future.

Scope of Viability
- High value for tax authorities in refund-risk screening.
- Applicable across sectors and regions.
- Supports targeted investigations (not automatic action).
- Can be extended to other tax compliance use cases.
- Potential for real-time or batch deployment.

Scalability
- Modular architecture for horizontal scaling (using Kubernetes).
- Cloud-ready design.
- Supports district/state-wise deployment.
- Efficient spatial preprocessing and graph analytics.
- Can handle millions of GSTINs with optimized pipelines.

Regulatory & Legal
- Must align with GST rules and refund risk provisions (e.g., Rule 86A).
- Use only permitted government and open data sources.
- Designed for human verification and due process.
- Maintain audit logs and explainability for accountability.

Data Availability
- Open geospatial data (Overture, WorldCover, OSM, Sentinel-1/2) available.
- Trade benchmarks from UN Comtrade and govt. statistics.
- Synthetic GST dataset for development and testing.
- Designed to integrate with real GST data in future.

Demo / MVP Readiness
- Working prototype with sample cases.
- Interactive dashboard with risk score and evidence dossier.
- Geospatial visualization on map.
- D1 (Price) and D4 (Network) initially, followed by D2 & D3.
- Can be demonstrated within hackathon timeline.

Future Potential
- Real-time risk alerts.
- Integration with government dashboards.
- AI-driven sector-specific models.
- Support for additional taxes and compliance use cases.
- Can evolve into a national-scale decision support platform.

Goal: A practical, scalable and transparent system that strengthens revenue protection through real-world evidence and AI — supporting, not replacing, human judgment.

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IMPACT AND BENEFITS

Smarter Detection • Stronger Compliance • A More Trustworthy Tax Ecosystem

IMPACTS

Citizens
Ensures genuine taxpayer funds are used properly, leading to fairer public services.

Businesses
Creates a level playing field by reducing fraudulent refund claims and unhealthy competition.

Government
Protects revenue, reduces financial leakages and enables data-driven policy making.

IMPACTS

Society
Strengthens trust in the tax system and promotes a more transparent and accountable ecosystem.

Tax Ecosystem
Builds a culture of compliance through deterrence and early detection of fraud networks.

Economy
Reduces illicit financial flows, improves market integrity and boosts economic stability.

BENEFITS

Operational Efficiency
Faster, data-driven screening reduces manual effort and investigation time.

Economic Benefit
Prevents wrongful refunds, saves public money and improves revenue realisation.

Risk Reduction
Early detection of suspicious entities, networks and claims reduces financial and reputational risk.

BENEFITS

Better Governance
Data-backed insights enable evidence-based policy decisions and targeted enforcement.

Scalability
Modular and cloud-ready design allows easy expansion to more HSN sectors, states and future use cases.

Trust & Confidence
Increases trust among genuine taxpayers and strengthens faith in the tax administration.

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RESEARCH AND REFERENCES

Strengthening Compliance • Detecting Fraud • Building a Transparent Tax Ecosystem

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KEY RESEARCH REFERENCES (ITC RADAR)

1 CBIC – GST Registration & Fake Entity Guidance
Guidelines on bogus registrations, non-functional entities and misuse of GST for fraudulent ITC.

2 GST Portal – GSTR-2B / Invoice Matching
Framework for supplier-reported invoices and recipient ITC reconciliation to ensure eligible and genuine claims.

3 DGGI – GST Intelligence & Fraud Detection
Use of data analytics, risk parameters and anti-evasion intelligence to identify suspicious transactions and fake ITC.

4 GST Laws, Rules & Notifications
Legal framework governing ITC eligibility, invoice requirements and penalties for fraudulent claims.

5 Academic Research – GST Fraud & Network Analytics
Studies on anomaly detection, graph/network analysis and explainable risk scoring for identifying fake invoices and ITC networks.

6 Geospatial / Premises Validation Research
Use of location data, satellite imagery and building capacity analysis to verify business premises and validate actual operations.

REAL-WORLD EVIDENCE — 2026 ITC/GST FRAUD CASES

15 JAN 2026 — CGST Delhi South
Detected ₹8.52 crore fraudulent ITC through bogus invoices worth ₹199.90 crore. Searches found closed/non-existent premises.

11 FEB 2026 — CGST Thane
Detected ₹172 crore fraudulent ITC involving goods valued ₹952 crore. Suppliers were non-existent or cancelled entities.

17 APR 2026 — CGST Delhi South
Detected more than ₹8 crore fraudulent ITC. Data analytics and backward supply-chain analysis identified missing underlying supplies.

20 APR 2026 — DGGI Ahmedabad
Unearthed approximately ₹1,825 crore GST refund fraud involving fraudulent ITC, dummy firms, layered invoices and fictitious/grossly exaggerated exports.

28 APR 2026 — CGST Delhi South
Detected ₹60.59 crore fraudulent ITC through bogus invoices valued ₹397.23 crore. Declared premises was a co-working space with no genuine commercial activity.

AUG 2026 — DGGI Lucknow
Detected approximately ₹185 crore GST evasion in an illegal pan masala/tobacco manufacturing operation. Coordinated searches seized 27 unregistered pouch-packing machines and 15.5 lakh pouches.

Official investigations/enforcement actions; cases may remain under investigation.

Complete Platform Prototype – Live Link
(To Be Added)

From Data to Evidence-Driven Tax Risk Detection